

Performance Testing

Date	15 July 2026
Team ID	SWUID2026217449
Project Name	Rising Waters - Flood Prediction ML
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing (Postman-style browser testing)
Type of Testing	Functional & Load Testing
Target Module / API	/predict endpoint (Flask route)
Test Environment	Local
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	High rainfall + high humidity input	1	5	Predicts High Flood Risk
2	Low rainfall + high cloud visibility input	1	5	Predicts Low Flood Risk
3	Boundary/edge case values	1	5	Model handles gracefully without crash
4	Repeated form submissions	5	10	App responds consistently under repeated use

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	< 1 second	Pass	Model prediction returns almost instantly
2	Response Time (Max)	< 5 seconds	< 1.5 seconds	Pass	No noticeable delay on local server
3	Throughput (Req/sec)		N/A	N/A	Not tested at scale (single-user local app)
4	Error Rate	< 1%	0%	Pass	No errors encountered during testing
5	CPU Utilization	< 80%	Low	Pass	Lightweight model, minimal CPU usage
6	Memory Utilization	< 80%	Low	Pass	Small model size, minimal memory usage

Step 4: Observations & Analysis

Key Findings:

- Predictions returned in under 1 second with no errors, and CPU/memory usage stayed low.

Bottlenecks Identified:

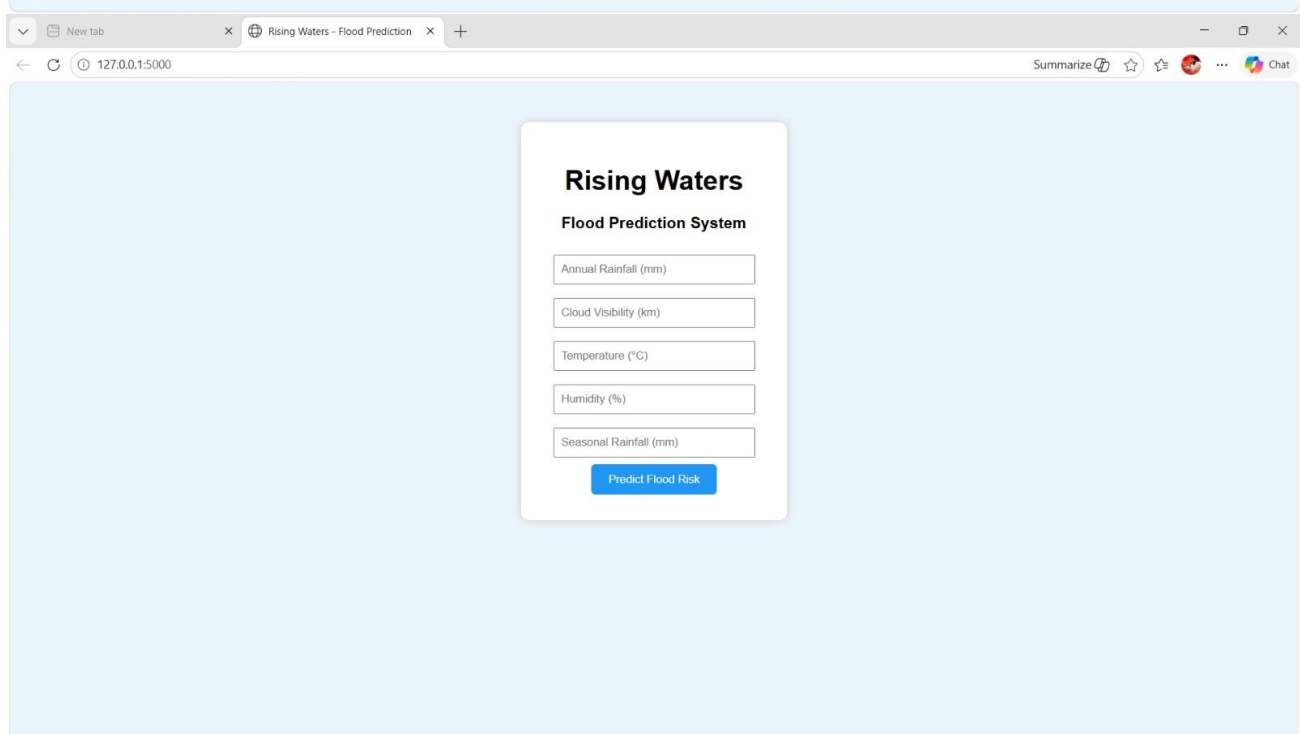
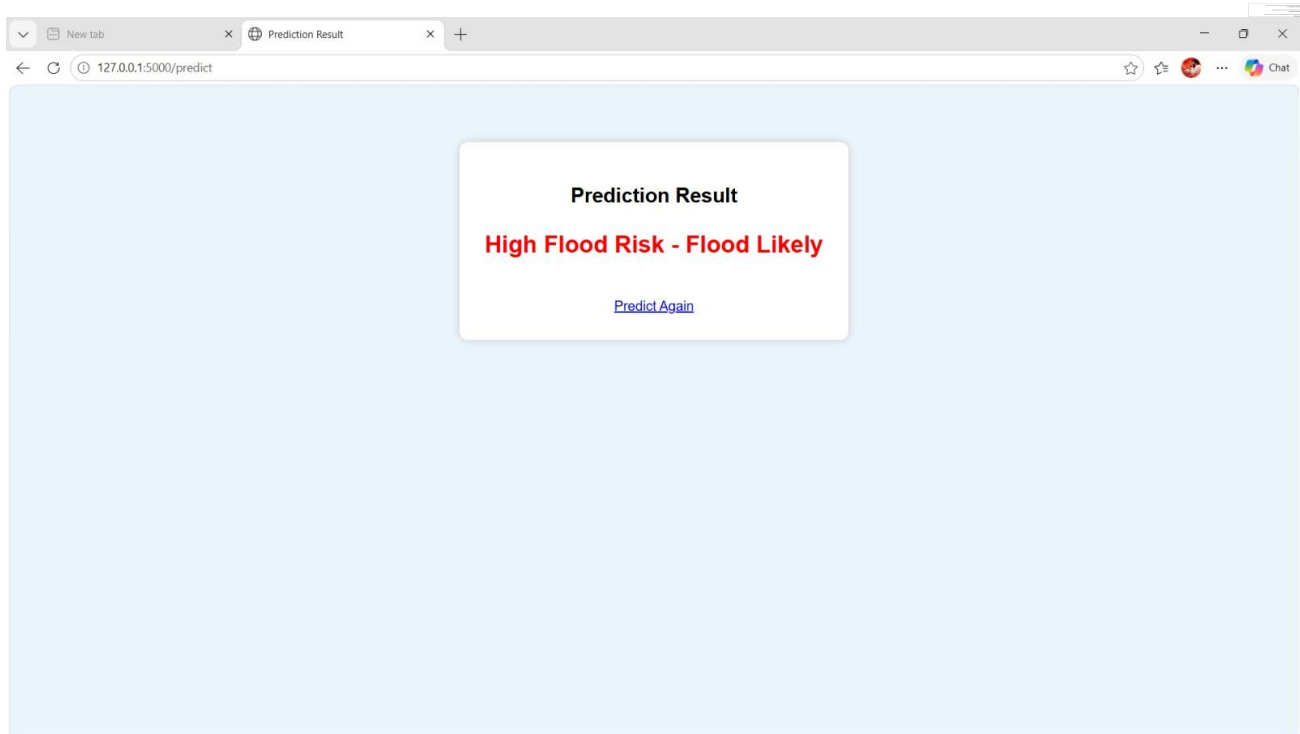
- No significant bottlenecks observed; the app ran fast and stable locally.

Optimization Steps Taken:

- Saved the trained model and scaler with pickle, and scaled inputs before prediction for faster, consistent results.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



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Rising Waters

Flood Prediction System

Predict Flood Risk